

Week 11: Electromechanical Energy Conversion

Force on a current, motional EMF, the linear machine as motor and generator, torque on a coil, magnetic pull, efficiency

In-depth notes covering all core material of this chapter: theory, engineering shortcuts, exam traps, 7 worked examples, and 8 practice problems with full solutions.

Based on: Rizzoni & Kearns, *Principles and Applications of Electrical Engineering*, 7th ed., §14.5

Inspired by: ELEC 275, Principles of Electrical Engineering, Concordia University

By **Vladislav Shkuratov**, made with Claude (AI). Independent study notes, not affiliated with Concordia University or any instructor.

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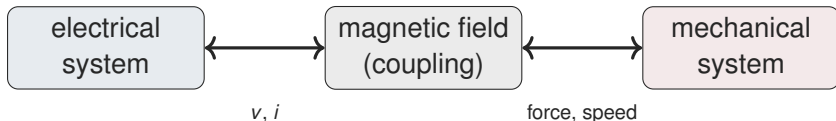
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How to use this deck

- **“Try it” slides:** the problem alone, with empty space. Do it first.
- **Next slide:** the clean solution, step by step.
- **Purple boxes:** engineering shortcuts, the fast way to see or solve things.
- **Orange boxes:** traps. This is where marks get lost on exams.
- Formulas are tagged **[on formula sheet]** or **[NOT on formula sheet]**, based on the sheet given in ELEC 275 exams.
- **End of deck:** final-check problems, then a full answer sheet.

The right side of each slide is left empty for your notes.

The big picture



Two effects do all the work:

1. **Force:** a current in a magnetic field feels a force. That's a **motor**.
2. **Induced voltage:** a conductor moving in a magnetic field gets a voltage. That's a **generator**.

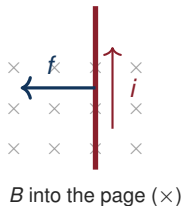
Every machine does both at once. Which one “wins” decides if it's motoring or generating.

Symbols for this week

Symbol	Unit	Meaning
B	T	magnetic flux density
ℓ	m	length of conductor inside the field
u	m/s	speed of the conductor
f	N	force
e	V	induced (motional) voltage, also called back EMF
T	N·m	torque
ω	rad/s	angular speed ($\omega = 2\pi n/60$ for n in rpm)
hp		horsepower: 1 hp = 746 W

Force on a current-carrying conductor

[NOT on formula sheet: memorize]



$$f = B i \ell$$

for a conductor perpendicular to B .

At an angle θ between the wire and B :

$$f = B i \ell \sin \theta.$$

Direction (right-hand rule): fingers along i , curl into B : thumb gives f . In vector form $\mathbf{f} = i \ell \times \mathbf{B}$.

Trap: only the wire **inside** the field counts in ℓ . A current parallel to B feels no force.

Where $f = Bi\ell$ comes from

A single charge q moving at velocity u through a field B feels a sideways force $F = quB$ (perpendicular to both). That's the basic magnetic force.

A current is just many charges moving along the wire. In a length ℓ , the total moving charge times its speed is $qu = i\ell$ (current = charge per second, and each charge covers ℓ in ℓ/u seconds). So

$$f = (qu)B = (i\ell)B = Bi\ell$$

Direction: perpendicular to both the wire and B . Right-hand rule: point your fingers along i , bend them toward B ; your thumb points along f .

Try it: Example 1, force on a conductor

A 0.5 m wire carries 10 A perpendicular to a field of 0.8 T.

Find the force. What if the wire were at 30° to the field?

your work here ↓

Example 1: solution

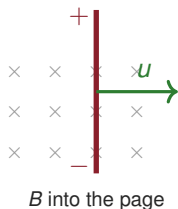
Formula $f = B i \ell \sin \theta$, with θ the angle between the wire and the field

Perpendicular $\theta = 90^\circ$, $\sin 90^\circ = 1$: $f = (0.8)(10)(0.5) = 4 \text{ N}$

At 30° only the part of the wire across the field counts:
 $f = 4 \sin 30^\circ = 4 \times 0.5 = 2 \text{ N}$

Motional EMF

[NOT on formula sheet: memorize]



$$e = B \ell u$$

A conductor moving at speed u across a field gets a voltage between its ends.

It's Faraday's law again: the moving bar sweeps area, so the flux through the circuit changes.

Lenz's law

The induced current always flows to **oppose** the motion that caused it. That opposing force is why a generator is hard to turn when it's loaded.

Where $e = B\ell u$ comes from

Put the moving bar on two rails, making a loop. At position x the loop encloses area ℓx , so the flux through it is $\Phi = B\ell x$.

The bar moves at $u = dx/dt$, so the flux changes at $\frac{d\Phi}{dt} = B\ell \frac{dx}{dt} = B\ell u$.

Faraday (one turn): $e = d\Phi/dt = B\ell u$.

Same result from charges: each free charge in the bar moves with it at u and feels quB along the bar. That pushes charges to one end, making a voltage $B\ell u$ between the ends.

Energy is conserved

Electrical power made: $ei = (B\ell u)i$. Mechanical power needed to push against the magnetic force: $fu = (Bi\ell)u$. They're **identical**: the magnetic field just converts one into the other.

Try it: Example 2, motional EMF

A 0.3 m bar moves at 10 m/s perpendicular to a 0.5 T field. What voltage is induced?

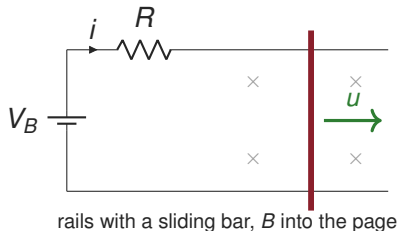
your work here ↓

Example 2: solution

Formula bar moving perpendicular to B : $e = B\ell u$

Result $(0.5 \text{ T})(0.3 \text{ m})(10 \text{ m/s}) = 1.5 \text{ V}$ ($\text{T}\cdot\text{m}\cdot\text{m/s} = \text{V}$)

The linear machine



A battery, a resistor, and a bar sliding on two rails.

KVL: $V_B = iR + e$, with $e = Bl u$

Force on bar: $f = Bil$

Steady speed: force balances the load, $Bil = f_{load}$.

Engineering shortcut: every machine is this circuit

Battery = supply, R = winding resistance, e = back EMF. The DC motor (week 12) is exactly this, just rotating: $V = I_a R_a + E_b$.

Try it: Example 3, the linear motor

$V_B = 12 \text{ V}$, $R = 0.2 \text{ } \Omega$, $B = 0.5 \text{ T}$, $\ell = 1 \text{ m}$.

- (a) Starting current and force (bar at rest).
- (b) No-load steady speed.
- (c) With a 10 N load force: current, back EMF, speed, and a power balance.

your work here ↓

Example 3: solution (1/2)

- (a) at rest $u = 0$ so $e = 0$: only R limits the current: $i = 12/0.2 = 60 \text{ A}$, $f = Bi\ell = (0.5)(60)(1) = 30 \text{ N}$
- (b) no load means no force is needed, so $i = 0$ and KVL gives $e = V_B$: $u = \frac{V_B}{B\ell} = \frac{12}{0.5} = 24 \text{ m/s}$
- (c) i steady speed: magnetic force balances the load, $Bi\ell = 10 \text{ N}$: $i = 10/0.5 = 20 \text{ A}$

Example 3: solution (2/2)

(c) e KVL: $e = V_B - iR = 12 - 20(0.2) = 8 \text{ V}$

(c) u $u = e/(B\ell) = 8/0.5 = 16 \text{ m/s}$

Balance in $V_B i = 240 \text{ W} = \text{resistor } i^2 R = 80 \text{ W} + \text{mechanical}$
 $f u = 160 \text{ W} \checkmark$

Why motors draw huge starting current

At start, $u = 0$ so $e = 0$: only R limits the current. R is small (to keep losses low), so the current is huge (60 A in Example 3 vs 20 A running).

As the bar speeds up, e grows and **fights** the supply, so the current drops. The back EMF is what makes a motor self-regulating: more load $\rightarrow$ slower $\rightarrow$ less $e \rightarrow$ more current $\rightarrow$ more force.

Trap

Real DC motors use a starting resistor (or a drive) to limit the starting current. Exam questions about “starting current” mean $e = 0$.

The linear machine as a generator

Push the bar **faster** than its no-load speed. Now $e > V_B$ and the current **reverses**: it flows into the battery (charging it).

	Motor	Generator
EMF vs supply	$e < V_B$	$e > V_B$
Current	out of the battery	into the battery
Force on bar	drives the motion	opposes the motion (you must push)
Energy flow	electrical $\rightarrow$ mechanical	mechanical $\rightarrow$ electrical

Shortcut: same machine, same equations. Only the sign of $V_B - e$ decides the mode.

Try it: Example 4, generating

The machine of Example 3 is pushed at a constant 30 m/s.

Find e , the current (and its direction), the force needed, and a power balance.

your work here ↓

Example 4: solution

EMF $e = B\ell u = (0.5)(1)(30) = 15 \text{ V}$, more than the 12 V battery: generating

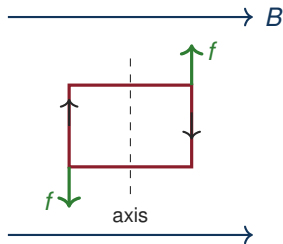
Current KVL: now e pushes current backward into the battery:
 $i = \frac{15 - 12}{0.2} = 15 \text{ A into the battery}$

Force the current now flows the other way, so the force **opposes** the motion (Lenz): $f = Bi\ell = 7.5 \text{ N}$

Balance pushed in $fu = 7.5 \times 30 = 225 \text{ W} =$ into the battery
 $12 \times 15 = 180 \text{ W} +$ lost in $R: 15^2 \times 0.2 = 45 \text{ W} \checkmark$

Torque on a coil: from linear to rotating

[NOT on formula sheet: memorize]



$$T = NBI A \sin \theta$$

N turns, area A , current I ; θ is the angle between B and the coil's normal.

Opposite sides of the coil get opposite forces: a **torque**. Maximum when the coil's plane is parallel to B ($\theta = 90^\circ$).

Without help, the torque reverses every half turn. A **commutator** (DC machines) or AC currents (AC machines) keep it pushing the same way.

Where $T = NBI\ell \sin \theta$ comes from

Take one rectangular turn: two sides of length ℓ cut across B , the other two sides have width w . Each long side feels $f = BI\ell$, one up and one down.

Torque = force $\times$ lever arm. Each side sits $w/2$ from the axis, but only the part of the force perpendicular to the arm turns the coil: arm $\times \sin \theta$. Two sides:

$$T = 2 \cdot BI\ell \cdot \frac{w}{2} \sin \theta = BI(\ell w) \sin \theta = BIA \sin \theta$$

N turns multiply it: $T = NBI\ell \sin \theta$, where θ is the angle between B and the coil's normal.

At $\theta = 0$ the forces pull straight outward (no turning). At $\theta = 90^\circ$ they're fully sideways: maximum torque.

Try it: Example 5, torque on a coil

A 100-turn coil, $10\text{ cm} \times 5\text{ cm}$, carries 2 A in a 0.5 T field.

Maximum torque?

your work here ↓

Example 5: solution

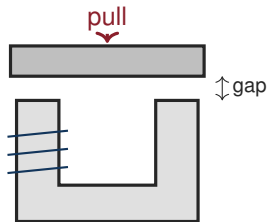
Area $A = 0.1 \times 0.05 = 0.005 \text{ m}^2$

T_{max} coil plane along B ($\theta = 90^\circ$, $\sin = 1$):

$$T = NBIA = (100)(0.5)(2)(0.005) = 0.5 \text{ N}\cdot\text{m}$$

Magnetic pull: relays, solenoids, lifting magnets

[NOT on formula sheet: memorize]



Energy stored in a gap pulls the pieces together (the magnetic “pressure”):

$$f = \frac{B^2 A}{2\mu_0} \text{ per gap face}$$

A = pole face area. A U-shaped magnet has **two** gaps: double it.

Why: closing the gap by dx releases stored energy $w A dx$, with $w = B^2/2\mu_0$ (week 10). Force = energy per distance.

Try it: Example 6, pull of a relay

A relay has one gap with pole area 4 cm^2 and $B = 1 \text{ T}$ in the gap. Find the pull.

your work here ↓

Example 6: solution

Formula one gap face: $f = \frac{B^2 A}{2\mu_0}$ (energy density $\times$ area)

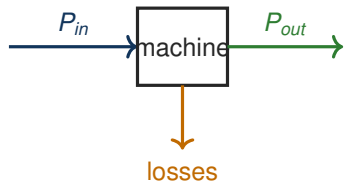
Area $4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

Result $\frac{(1)^2 \times 4 \times 10^{-4}}{2 \times 4\pi \times 10^{-7}} = \frac{4 \times 10^{-4}}{2.51 \times 10^{-6}} = 159 \text{ N}$

About 16 kg of pull from a $2 \text{ cm} \times 2 \text{ cm}$ face. $f \propto B^2$: halve B and the force drops to a quarter.

Energy balance and efficiency

[NOT on formula sheet: memorize]



$$\eta = \frac{P_{out}}{P_{in}} = \frac{P_{out}}{P_{out} + P_{losses}}$$

Losses: copper ($I^2 R$), core (hysteresis, eddy), friction and windage.

Mechanical power

Linear: $P = f u$. Rotating: $P = T \omega$ (with ω in rad/s). Motors are rated by **output** (shaft) power, often in hp.

Try it: Example 7, efficiency

A motor delivers 5 kW at the shaft with 88% efficiency. Find the electrical input and the losses.

your work here ↓

Example 7: solution

Rule motors are rated by **output**; efficiency = P_{out}/P_{in} , so

$$P_{in} = P_{out}/\eta$$

Input $5/0.88 =$ 5.68 kW

Losses input – output = $5.68 - 5 =$ 682 W

Trap: input = rated $\div \eta$, not $\times \eta$ (the input is always the bigger number).

Traps

Check these before you box an answer

- $f = B\ell v$ and $e = B\ell u$ assume perpendicular directions. Otherwise add $\sin \theta$.
- Starting ($u = 0$): $e = 0$, the current is only limited by R .
- No load: $i = 0$, so $e = V_B$ (top speed).
- Motor if $e < V_B$, generator if $e > V_B$.
- Rated power is output. Input = P_{out}/η .
- rpm to rad/s: $\omega = 2\pi n/60$.

Exam-format traps

Always do a power balance

Input = output + losses. It takes 30 seconds and catches sign and unit errors.

State the mode

Say “motoring” or “generating” and the current’s direction. It’s often worth a mark on its own.

Summary

Force

$f = Bil$ (right-hand rule)

Motional EMF

$e = B\ell u$; Lenz: it opposes the motion

Linear machine

$V_B = iR + e$; start: $e = 0$; no load: $i = 0$

Modes

motor $e < V_B$; generator $e > V_B$

Coil torque

$T = NBIA \sin \theta$

Magnetic pull

$f = B^2 A / (2\mu_0)$ per gap face

Power

$P = fu = T\omega$; $\eta = P_{out}/P_{in}$; 1 hp = 746 W

Final check

8 problems. One per slide, with space. No looking back.

If one takes more than 10 min, go back to that section. The answer sheet comes after the last problem.

P1: force

A 0.2 m wire carries 5 A in a 1.2 T field at 30° to the field. Find the force.

your work here ↓

P2: EMF

A 0.5 m bar moves at 20 m/s across a 0.4 T field. Induced voltage?

your work here ↓

P3: linear motor

$V_B = 24 \text{ V}$, $R = 0.5 \text{ } \Omega$, $B = 0.8 \text{ T}$, $\ell = 0.5 \text{ m}$.

(a) No-load speed (b) Starting current and force (c) Speed with an 8 N load

your work here ↓

P4: generator

The P3 machine is pushed at 70 m/s. Find the current, the force needed, and a power balance.

your work here ↓

P5: coil torque

A 50-turn coil of area 0.02 m^2 carries 3 A in a 0.4 T field.

Maximum torque, and the torque at $\theta = 30^\circ$.

your work here ↓

P6: lifting magnet

A U-shaped electromagnet has two pole faces of 6 cm^2 each, with $B = 0.8 \text{ T}$ in each gap. Total pull?

your work here ↓

P7: efficiency

A motor takes 12 kW and loses 1.2 kW. Efficiency, and output in hp?

your work here ↓

P8: exam style

A linear machine: $B = 0.6$ T, $\ell = 0.8$ m, $V_B = 48$ V, $R = 0.4$ Ω , load force 12 N.

Find the current, the back EMF, the steady speed and the efficiency.

your work here ↓

Answer sheet

Full worked solutions to P1–P8.

P1: solution

Formula $f = B i \ell \sin \theta$

Result $(1.2)(5)(0.2) \sin 30^\circ = 1.2 \times 0.5 = 0.6 \text{ N}$

P2: solution

Formula $e = B\ell u$

Result $(0.4)(0.5)(20) = 4 \text{ V}$

P3: solution

$B\ell$ $0.8 \times 0.5 = 0.4$ (appears in both $e = B\ell u$ and $f = B\ell i$)

(a) no load: $i = 0$, $e = V_B$: $u = 24/0.4 = 60 \text{ m/s}$

(b) at rest $e = 0$: $i = 24/0.5 = 48 \text{ A}$, $f = 0.4 \times 48 = 19.2 \text{ N}$

(c) i force balances the 8 N load: $i = 8/0.4 = 20 \text{ A}$

(c) u $e = 24 - 20(0.5) = 14 \text{ V}$, $u = 14/0.4 = 35 \text{ m/s}$

P4: solution

EMF $e = 0.4 \times 70 = 28 \text{ V}$, more than $V_B = 24 \text{ V}$: generating

Current $(28 - 24)/0.5 = 8 \text{ A into the battery}$

Force $0.4 \times 8 = 3.2 \text{ N}$, opposing the motion

Balance pushed in $3.2 \times 70 = 224 \text{ W} = \text{battery } 24 \times 8 = 192 \text{ W} +$
loss $8^2 \times 0.5 = 32 \text{ W} \checkmark$

P5: solution

$$T_{max} \sin \theta = 1: (50)(0.4)(3)(0.02) = 1.2 \text{ N}\cdot\text{m}$$

$$\text{At } 30^\circ \quad 1.2 \sin 30^\circ = 0.6 \text{ N}\cdot\text{m}$$

P6: solution

Per face $\frac{B^2 A}{2\mu_0} = \frac{0.8^2 \times 6 \times 10^{-4}}{2(4\pi \times 10^{-7})} = 152.8 \text{ N}$

Total a U-shaped magnet has **two** pole faces (two gaps), both pulling: 306 N

P7: solution

$$\eta \text{ output} = 12 - 1.2 = 10.8 \text{ kW: } 10.8/12 = 90\%$$

$$\text{hp } 10\,800/746 = 14.5 \text{ hp}$$

P8: solution

$$B\ell \quad 0.6 \times 0.8 = 0.48$$

Current force balances the 12 N load: $i = 12/0.48 = 25 \text{ A}$

Back EMF KVL: $48 - 25(0.4) = 38 \text{ V}$

Speed $u = e/(B\ell) = 38/0.48 = 79.2 \text{ m/s}$

Efficiency mechanical out / electrical in: $\frac{12 \times 79.2}{48 \times 25} = \frac{950}{1200} =$
 79.2% (the rest, 250 W, is $i^2 R$)